

A Real-Time Hybrid Agentic AI Browser Extension for Multilingual Legal Risk Detection, Clause Analysis, and Summarization

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Abstract—Terms of Service agreements and Privacy Policies are often long and written in legal language. This makes them hard to understand for a lot of users [1], [9]. Some systems only work on Natural Language Processing tasks while others use Large Language Models a lot [6], [7]. Large Language Models are pretty useful because they can do things but using them can also cause some problems like answers that do not always make sense, it takes a lot of computer power to run them, and it is hard to use them in real life situations, with Terms of Service agreements and Privacy Policies [9], [12]. Real-time legal document analysis is supported through a hybrid AI-based framework. The system operates through a browser extension while the backend processing is handled by a FastAPI server [10], [15]. The analysis process follows a deterministic pipeline for handling legal documents. Zero-shot clause classification, transformer-based models, and rule-based fallback methods are used together to improve clause detection [2], [6]. A weighted risk scoring process is used to generate stable results, even when computational resources are limited [4], [13]. LLM refinement may be used to improve explanations and support multilingual responses when needed [12]. The main analysis process continues independently, even when this refinement step is not used [13]. Users get inline warnings when they enter sensitive information in forms. These warnings show privacy risks as they happen, not after they are done [10], [15]. Methods that only use LLMs can give different answers but our framework gives more reliable results and responds quicker [9], [13]. The legal document analysis process is integrated with an interface that helps users to identify and understand privacy-related risks while interacting with the system [10], [15].

Keywords: Privacy Policy Analysis; Clause Detection; Risk Assessment; Hybrid AI; Real-Time Systems; Browser Extension; Legal NLP

I. INTRODUCTION

Over time, the approaches for legal document understanding have been various, from classical machine learning methods to more recent transformer-based and generative AI models. In earlier approaches, legal documents were processed using rule-based extraction and hand-crafted features [2]. The introduction of transformer models such as BERT later improved clause classification and contextual understanding [8]. LLMs are now often used to analyse legal texts such as Terms of Service (ToS) and Privacy Policies [6], [7]. They are mainly

used to generate readable summaries and improve understanding of legal content [12]. Terms of Service and Privacy Policy documents are legally important but many users may still find them difficult to understand because they are usually long and written in legal language [1]. As a result agreements may be accepted without fully recognizing possible privacy risks [9]. LLMs can help make these documents easier to understand, allowing users to better understand what they are agreeing to [6], [15]. Existing automated systems also face several limitations. Many operate as monolithic pipelines, provide limited multilingual support, and do not offer contextual inline warnings during user data entry [10], [15]. Risk assessment is not always detailed, and outputs may sometimes lack a clear structure for legal or privacy-related use [9], [13]. Because of these limitations, a hybrid approach becomes necessary for legal document analysis [13]. Deterministic processing is combined with LLM-based semantic reasoning, while real-time browser deployment is also supported [10], [15]. Privacy-related risks may be shown to users during interaction through warnings at the time of data entry, rather than relying only on policy documents [10], [15]. Multiple language support is also available. This allows users to view information in a familiar language helping them understand the content more clearly [15]. For real-time legal clause detection and privacy policy analysis, a hybrid AI-based system is used. The system operates through a browser extension, while backend processing is managed by a FastAPI server [10], [15]. Several components are included in the framework. A deterministic core pipeline is used for clause detection and risk scoring, while LLMs are applied selectively in ambiguous cases to support a semi-agentic execution flow [2], [13]. Contextual inline warnings provide field-level privacy alerts during data entry [10], [15]. We also enable multi-lingual summarization so users with different language backgrounds can understand information more easily [15].

II. LITERATURE REVIEW

Privacy Policies and Terms of Service (ToS) documents are often difficult for non-expert users to understand because they are usually long and written in legal language [1], [9]. In many

cases these documents are written mainly to meet regulatory requirements rather than help users understand the content. Many users struggle to understand privacy policies as intended. Users may agree without fully understanding what they are accepting. Users may sometimes accept policies without fully recognizing privacy risks such as data sharing, tracking or misuse of personal information [1], [15]. Researchers began exploring automated legal document analysis as privacy concerns continued to grow. Earlier approaches to privacy policy analysis mainly used rule-based systems and classical machine learning techniques [2]. Some tools used predefined rules to identify privacy risks. Others relied on visual summaries to explain important policy practices in a simpler form. These methods are generally easy to interpret and work efficiently, but adapting to changing legal language remains difficult [2]. Classical machine learning models improved generalization, though they often required extensive feature engineering [7]. Transformer-based models such as BERT were later used for clause classification tasks and showed better contextual understanding [8]. Even so, these models are usually limited to specific tasks, depend heavily on labeled datasets, and may not always be easy for end users to interpret [7], [8]. Large Language Models (LLMs) are being used more frequently in legal document analysis [6], [7]. LLM-based systems are often used to generate readable summaries and help interpret complex legal content [6], [12]. PRISMe [10] introduces a browser extension that summarizes policies and supports user queries to help users understand policies better. Smart Privacy Policy Assistant [15] uses LLMs to extract clauses and provide real-time risk warnings within a browsing environment. PolicyGPT [6] is another system that automates policy analysis using generative models. These approaches improve accessibility and user interaction, but they often depend heavily on LLM outputs. This may introduce inconsistencies, reduce deterministic validation, and raise concerns about reliability in legal and privacy-related applications [9], [12]. Even with these developments, some limitations still remain. Rule-based and classical machine learning methods are often less flexible in understanding language [2]. Transformer-based models are usually limited to specific tasks [8]. LLM-centric systems also depend heavily on generated responses without rule-based verification which may reduce reliability in important privacy-related decisions [12], [13]. Most existing systems also do not provide real-time, context-aware assistance during user interaction, such as contextual inline form warnings [10], [15]. Limited work currently combines deterministic rule-based analysis with controlled LLM refinement in a single architecture [13]. A hybrid framework that combines reliable performance, better understanding, and real-time support is still limited. This points to an important gap in existing research. Because of these limitations, a hybrid system becomes necessary to perform reliable analysis while using LLMs in a controlled way to improve user understanding [13], [14]. Table I compares existing approaches used for privacy policy analysis, including their methods, advantages, and limitations. Existing approaches shown in Table II include rule-based, ma-

chine learning, transformer-based and LLM-based techniques. Rule-based and classical methods are easier to understand. However, handling complex legal language is still difficult [2]. Transformer-based approaches improve classification accuracy. At the same time, they are often limited to specific tasks and usually require labeled datasets [7], [8]. LLM-based systems help improve usability and user interaction [6], [12]. Reliability issues may still occur, and validation is not always consistent [9], [12]. These limitations make a hybrid approach more useful for combining structured analysis with flexible understanding and real time user interaction [13], [15].

As shown in Table II, existing approaches vary across rule-based, machine learning, transformer-based, and LLM-based techniques. While rule-based and classical methods provide interpretability, they lack adaptability to complex legal language. Transformer-based approaches improve classification accuracy but are often limited to specific tasks and require labeled datasets. LLM-based systems enhance usability and interaction but may suffer from reliability issues and lack deterministic validation. These limitations highlight the need for a hybrid approach that combines reliable structured analysis with flexible semantic reasoning and real-time user interaction.

III. METHODOLOGY

A. Framework

We designed the framework to automatically analyze Terms of Service and Privacy Policies and make them easier for users to understand [1], [15]. We used a rule-based pipeline along with an optional LLM layer to make the results easier to understand [2], [13]. To keep the system reliable, the model is used only to improve explanations when needed. We use the refinement layer only for making explanations clearer where certain clauses are difficult to understand [12]. The LLM helps explain complex clauses, the final output generated by the rule-based system stays the same [13]. For easier access, we integrated it into a browser extension that works during normal web browsing [10], [15]. The system has two main components: the browser extension interface and the backend processing module. The browser extension collects legal text from websites and sends it to the backend for analysis. Users can then view summaries, highlighted clauses, and risk warnings [10]. The extension shows results in real time, helping users understand policies more easily [15]. The backend is implemented using FastAPI and handles tasks such as text preprocessing, clause detection, risk scoring, and summary generation. It also supports optional LLM-based refinement and sends the processed results back to the frontend in JSON format [15]. To evaluate the system, we used legal documents such as privacy policies and Terms of Service from publicly accessible websites. Privacy policies from different websites are written in different formats and legal styles [7]. Because of this variation, we used zero-shot classification to handle unseen documents without depending completely on labeled datasets [6], [7]. This helps the system analyze new privacy policies without depending too much on large labeled datasets making the system more practical for real-world use. Python

TABLE I
COMPARISON OF EXISTING APPROACHES FOR PRIVACY POLICY ANALYSIS

Paper	Method Used	Key Contribution	Limitations
Polisis (Apthorpe et al., 2019)	Deep Learning (CNN)	Automated classification of privacy policy segments	Limited handling of novel clauses
CUAD (Hendrycks et al., 2021)	Dataset + ML	Large annotated dataset for legal clause detection	Focused on contracts, not user understanding
PoliGraph (Li et al., 2018)	Knowledge Graph	Structured representation of policies	Complex and hard to maintain
Privacy at Scale (Xie et al., 2025)	LLM-based	Large-scale analysis of privacy practices	No real-time user interface
Entailment Framework (Wu et al., 2023)	LLM (NLI)	Clause classification using entailment reasoning	Sensitive to input formulation
Word-GDPR (van Engelenburg et al., 2020)	Transformer + Annotation	Fine-grained GDPR annotation	Limited to regulatory context
PolicyGPT (Chen et al., 2024)	LLM (GPT)	Automated policy summarization	Risk of hallucination
BERT-GDPR (Mody et al., 2021)	Transformer (BERT)	High-accuracy GDPR clause classification	Requires labeled data
PRISMe (Freiberger et al., 2025)	LLM-based	Real-time browser extension for policy understanding	Lacks deterministic validation
LLM Explainability (Lee et al., 2024)	LLM + Explanation	Improves interpretability of outputs	Increased system complexity
Smart Privacy Policy Assistant (Kalvakuntla et al., 2026)	LLM-based	Real-time clause detection and risk warnings	Fully dependent on LLM
Privacy Bird (Cranor, 2006)	Rule-based	Early privacy risk detection tool	Static and outdated
Privacy Nutrition Label (Kelley et al., 2009)	Visualization	Simplified presentation of privacy practices	Not interactive
ToS;DR (Hansen et al., 2014)	Crowdsourced	Community-driven policy summaries	Inconsistent coverage
Proposed System (This Work)	Hybrid (Rule + LLM)	Deterministic analysis with controlled LLM refinement and real-time interaction	Designed to address above limitations

and FastAPI were used to develop the backend of the system, while JavaScript and Chrome Extension APIs were used for the browser extension interface [10]. The framework also supports LLM APIs such as GPT and Gemini for improving explanations and handling complex clauses when needed [12], [15]. The frontend and backend communicate using JSON format. The system runs on a computer with at least 8 GB RAM, 10 GB of free storage, and an Intel i5 processor or a similar setup. Internet access is needed for LLM APIs and related online services.

B. Algorithm

1) Software Requirements:

Python (Backend Development) FastAPI (API Framework)
JavaScript (Frontend Development) Chrome Extension APIs
(Browser Integration) LLM APIs (e.g., GPT, Gemini) JSON
(Data Exchange Format)

2) Hardware Requirements:

Processor: Intel i5 or equivalent RAM: 8 GB or higher Storage: Minimum 10 GB available space Internet Connectivity: Required for API-based LLM access

Algorithms and Processing Techniques:

The complete workflow is divided into seven stages:

1) Text Preprocessing Algorithm: Text Preprocessing Algorithm: The preprocessing step removes unwanted characters and extra spaces from the input text. After cleaning, the text is broken into smaller sentences so it becomes easier to work with [1], [7].

2) Sentence Relevance Selection Algorithm: Relevant Sentence Selection Algorithm: First, unnecessary characters and

extra spaces are removed from the text. It is then split into smaller sentences so the content becomes easier to process [7].

3) Zero-Shot Clause Classification Algorithm: Zero-Shot Clause Classification Algorithm: Here, legal sentences are grouped into categories such as data sharing, tracking, liability, and user rights using a zero-shot model [6]. This approach reduces the need for large labeled training datasets and also helps handle new privacy policies [6], [7].

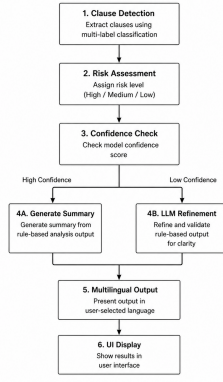
4) Rule-Based Keyword Fallback Algorithm: Rule Based Keyword Fallback Algorithm: In some situations the main model may fail to identify a clause correctly or may generate uncertain results. To handle such cases the framework applies a keyword based fallback approach [2]. This method searches for predefined privacy related keywords and patterns so that important clauses are not left undetected.

5) Clause Consolidation Algorithm: Clause Consolidation Algorithm: Many privacy policies contain clauses that express similar ideas using different wording. Instead of displaying repeated information the framework combines related clauses and keeps the most relevant one [1]. As a result the final output becomes more organized and easier to review.

6) Risk Scoring and Categorization Algorithm: Risk Scoring and Categorization Algorithm: Once the clauses are extracted the framework analyzes the possible privacy risk linked with each one [4]. Depending on the sensitivity and impact of the content the clauses are grouped into categories such as Low, Medium, or High risk [15].

7) LLM-Based Validation and Refinement Algorithm (Optional): The optional LLM module is mainly used when the

FIG. 1: WORKFLOW / PROCESSING FLOW



framework is not fully confident about a result [9], [12]. If the output already appears accurate no further refinement is carried out. This helps improve uncertain results while avoiding unnecessary processing. LLM helps improve explanations while keeping the original rule-based results unchanged [13]. Even if the LLM is unavailable, the system can still continue working using the rule-based analysis [13].

C. System Workflow

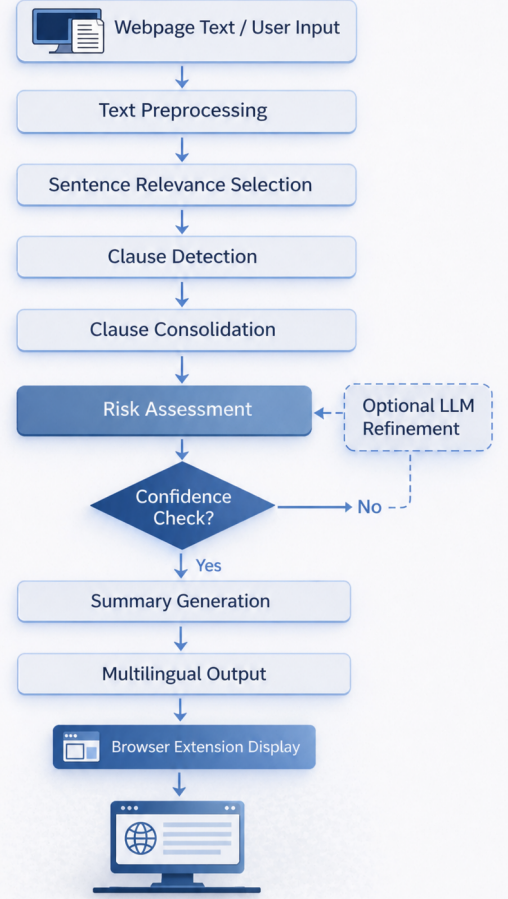
Fig 1 shows the overall workflow of the framework.

First, legal text is taken either from webpages or directly from user input. The text is then cleaned by removing extra spaces and unwanted characters [1], [7]. After that the framework looks for important legal sentences in the document. Important legal clauses are detected and similar clauses are then grouped together to reduce repeated results [2]. After that privacy risks are evaluated using rule based scoring methods [4]. The detected clauses are then used to generate a summary [1]. When the model gives uncertain results the optional LLM helps provide a clearer explanation [12], [13]. The browser extension displays the analysis results including summaries and privacy warnings so users can check important privacy related information while browsing [10], [15]. It also supports different languages so users can read summaries and privacy warnings more comfortably [15]. Once a language is selected the results are shown in the same language. Multiple language support is provided using translation templates and optional LLM based refinement [12], [15]. The core processing steps remain the same across different languages which helps keep the output consistent [7]. Once processing is completed the browser extension displays the results in the language selected by the user. When available LLM also helps make the explanations easier to read and more natural [12]. The extension can still work even when LLM support is unavailable. In such cases template based translation is used to display the results in a simple and understandable format [13]. This keeps the results similar across different languages and allows the extension to continue working properly.

D. MultiLingual Section

The proposed system includes multilingual support so users can access the platform in their preferred language [15]. Users

FIG. 2



can select their preferred language directly from the browser extension interface; the system then displays summaries, risk levels, key concerns, and clause explanations in the selected language [10], [15]. The multilingual functionality is implemented using a hybrid localization approach. The main analysis process works independently of language and uses normalized text for processing. This helps maintain consistent clause detection and risk analysis [7]. The output layer is then adapted using structured templates and localization mappings, while the optional LLM refinement layer improves fluency and contextual clarity when available [12]. Importantly, the system maintains functionality even in the absence of LLM support. In such cases, template-based translation ensures that outputs remain structured and understandable. This design guarantees robustness, consistency, and scalability across multiple languages, making the system suitable for real-world deployment in diverse linguistic environments [15].

IV. SYSTEM DESIGN / ARCHITECTURE

The overall workflow of the proposed system is illustrated in Fig.2.

The system takes webpage legal text or user input as input and processes it through a multi-stage pipeline. Initially, text

preprocessing is performed to clean and normalize the input. Relevant sentences are then selected using a heuristic-based relevance mechanism.

The selected sentences undergo clause detection followed by clause consolidation to group similar legal statements. A deterministic risk assessment module evaluates the severity of detected clauses.

A confidence-based decision mechanism determines whether the output should be refined. If the confidence is below a predefined threshold, the optional LLM refinement module is invoked to improve explanation quality. Otherwise, the deterministic output is retained.

Finally, the system generates a structured summary, supports multilingual output, and displays the results through the browser extension interface.

V. RESULTS AND DISCUSSION

A. Experimental Observations

The proposed system was evaluated qualitatively by observing its functionality, runtime behavior, and performance against existing privacy policy analysis tools. Instead of focusing only on classification accuracy, the evaluation also considers real-time usability, interpretability, and system robustness.

The system successfully performs clause-level detection, risk scoring, and summary generation across diverse real-world privacy policies. The rule-based core produces stable and deterministic outputs, while the optional LLM refinement layer improves explanation clarity without affecting system stability. The system also continues to function correctly even when the LLM layer is unavailable.

During practical testing, the system demonstrated:

- Accurate clause categorization using zero-shot classification with rule-based fallback
- Stable risk scoring through deterministic evaluation
- Clear and structured summaries generated independently of LLM support
- Better explanation clarity when optional LLM refinement is enabled

Additionally, the system also supports multilingual output, allowing summaries, risk indicators, and clause explanations to be displayed in the user's preferred language. This improves accessibility for non-English-speaking users while maintaining consistency in analysis results.

Overall, the observations indicate that the system maintains a balanced combination of efficiency, interpretability, and semantic understanding.

B. Comparative Analysis with Existing Systems

For evaluation purposes, the proposed framework was compared with two similar systems: PRISMe and Smart Privacy Policy Assistant.

1) *Comparison with PRISMe*: Comparison with PRISMe: PRISMe [10] is an LLM based browser extension developed to provide interactive privacy policy analysis along with user friendly explanations. It includes features such as a dashboard and conversational interface that help users understand privacy

related content more easily. However, the system depends heavily on LLM driven processing for clause interpretation and evaluations. Some earlier studies have also mentioned that systems relying completely on LLM generated outputs can sometimes produce inconsistent responses since the generated explanations may vary from one case to another and may include inaccurate information [9], [10].

In comparison, the proposed framework follows a more deterministic processing pipeline. Clause detection and risk scoring are mainly controlled through rule based logic, while the LLM component is used only as an optional refinement layer instead of direct decision making [2], [13]. Because of this, the generated outputs remain more stable, interpretable, and consistent during evaluation. Overall, the proposed system improves upon PRISMe [10] by reducing hallucination related issues while also providing more reliable and explainable results.

Comparison with Smart Privacy Policy Assistant: Smart Privacy Policy Assistant [15] is another LLM based system developed for privacy policy analysis. It performs tasks like clause extraction, risk scoring, and explanation generation by following a predefined structured format for organizing and processing policy related information. The system also supports real time usage and provides contextual warnings that can help users better understand privacy related concerns while interacting with policies. Even though the system offers understandable risk scoring most of its analysis still depends on LLM generated outputs for both classification and explanation. In some situations especially when the policy language becomes more complex or legal in nature such approaches may produce responses that are not always well structured [9], [12]. The developed framework works differently because it combines multiple techniques instead of depending entirely on the LLM. Safety related decisions are mainly handled using deterministic rules [2], [4], while zero-shot classification is used to improve semantic understanding during clause analysis [6], [7]. The LLM component is only used for refinement and validation purposes when required [12], [13]. Due to this hybrid design, the framework is able to generate outputs that are more structured and consistent during testing [13], [14]. It also provides better stability in real time environments while keeping the use of generative models more controlled [13]. Table II shows a broader comparison of existing privacy policy analysis systems, including their methodologies, contributions, and limitations.

Table II presents a comparison of existing approaches for privacy policy analysis, highlighting their methodologies, contributions, and limitations.

C. Discussion

The comparative analysis shows that the developed framework attempts to overcome several limitations found in existing systems by following a more controlled hybrid approach [13], [14]. Many LLM focused systems can sometimes produce inconsistent outputs or explanations that are difficult to verify [9], [12]. To avoid this issue the framework keeps

TABLE II
COMPARATIVE ANALYSIS OF SYSTEMS

Feature	Proposed System	PRISMe	Smart Assistant
Core Approach	Hybrid (Rule + Zero-shot + LLM)	LLM-driven	LLM-driven
Decision Control	Deterministic (Rule-based)	LLM-based	LLM-based
LLM Role	Optional refinement only	Primary engine	Primary engine
Reliability	High	Medium (LLM variability)	Medium
Interpretability	High	Moderate	Moderate
Real-Time Support	Yes	Yes	Yes
Inline Warnings	Yes (Proposed feature)	No	Limited
Multilingual Support	Yes (Output-level)	LLM-dependent	LLM-dependent
Dependency on LLM	Low	High	High

important decisions deterministic and easier to track during analysis [2], [13]. One of the main advantages of the system is the way different components are assigned separate responsibilities. Rule based methods help maintain stability and correctness during risk related decisions [2], [4] while zero-shot classification improves flexibility during clause analysis [6], [7]. The LLM module is only used for refinement and improving explanation quality instead of handling the core decision making process directly [12], [13]. Another feature is the use of contextual inline form warnings. Instead of limiting the system to static document analysis, the framework can also provide real time guidance while users enter information into forms [10], [15]. This makes the interaction more practical and user oriented compared to many existing approaches. The multilingual support further improves accessibility by allowing users to read summaries and explanations in their preferred language [15]. At the same time the core analysis process remains stable because the system does not rely completely on LLM generated outputs for critical operations [13]. Overall, the results shows that the proposed system achieves a balance between accuracy, interpretability, real time usability, and system robustness [13], [14], [15].

VI. IMPLEMENTATION DETAILS

A. Tools and Technologies Used

The implementation combines frontend, backend, and machine learning components that work together to analyze privacy policies in real time [10], [15]. The frontend handles user interaction through the browser extension, the backend manages processing and communication between services and the machine learning models are used to understand and classify privacy related content automatically. For the browser side functionality JavaScript was used to build the extension responsible for collecting webpage content and sending it to the backend for further processing [10]. On the backend side, Python along with the FastAPI framework was used to handle API requests and document processing tasks [15]. FastAPI was mainly chosen because it is lightweight, simple to work

with and efficient when dealing with real time operations. For clause analysis and classification transformer based NLP models were integrated into the framework [8]. These models were mainly used for zero-shot classification so that legal clauses could be categorized without requiring extensive task specific training [6], [7]. Along with this, rule based methods were also added in some critical parts of the pipeline especially for fallback clause detection and risk scoring since these components required more stable and predictable behavior [2], [4]. The framework also supports optional integration with LLM APIs. These APIs are mainly used for improving explanation quality and generating multilingual outputs whenever needed [12], [15]. However, the core analysis pipeline does not rely completely on the LLM which helps maintain consistency in the final results [13].

B. Libraries and Frameworks

FastAPI was used for the backend. It kept the communication between different modules easy to manage and did not make things more complicated than they needed to be [15]. HuggingFace Transformers handled the zero-shot classification for spotting privacy-related clauses [6], [9]. Training something from scratch for that would have taken way more time and effort than it was worth. Scikit-learn, NumPy, and Pandas covered most of the data side anything from basic processing to tasks that needed a little more work analytically [7]. The Requests library was used when the system needed to connect with external LLM APIs [12]. JavaScript and the Chrome Extension APIs handled everything on the browser side, interacting with webpages and pulling out policy-related content [10]. Using all of these together meant the project stayed organized and the different pieces of the hybrid AI pipeline did not end up conflicting with each other [13].

C. System Setup and Execution

The system operates as a real time pipeline integrated into the user's browsing environment [10], [15]. The workflow begins when the user activates the browser extension on a webpage containing Terms of Service or Privacy Policy

content. The browser extension collects text from the webpage and sends it to the FastAPI backend using an API call. Along with the extracted content information such as the webpage link and selected language is also passed to the backend [10]. Before the analysis starts the backend processes the content by removing extra text, splitting the document into smaller sentences and identifying the sections that appear more relevant for processing [1], [7]. After preprocessing, the text is forwarded to the clause detection module where zero-shot transformer models are used to identify and classify sentences into predefined legal categories [6], [9]. If the model is unable to generate predictions in certain situations the system switches to a rule based fallback method so that the overall processing can continue without interruption [2]. After the clauses are detected similar or repeated categories are combined together to avoid duplicate results. While merging them the system keeps the highest severity level and confidence value associated with those clauses [1]. The final consolidated output is then passed to a rule based risk analysis module which calculates the overall risk score, assigns a risk level such as Low, Medium, or High, and generates important concerns along with suggested recommendations [4], [15]. Based on the generated results the system creates a template based summary containing the main findings and risk related information [1]. When the confidence score falls below a predefined limit an optional LLM refinement layer is used to verify the output and improve the clarity of explanations [9], [12]. However, the refinement step does not replace the deterministic decisions already produced by the framework [13]. After all processing steps are completed the final results are converted into a structured JSON format and sent back to the browser extension [15]. The extension then presents the output in the user interface in the form of risk banners, summaries, clause wise explanations and multilingual results based on the language selected by the user [10], [15].

VII. CONCLUSION

- Summary of work

This work presented a hybrid AI-based framework for analyzing Terms of Service and Privacy Policy documents in real time [2], [6], [13]. The system combines rule based methods, zero-shot transformer models, contextual inline warnings and an optional LLM refinement layer within a modular processing pipeline. A browser extension together with a FastAPI backend was used to support smooth real time analysis and display privacy related insights directly inside the user's browsing environment [10], [15]. The framework was designed in a way that important decisions remain deterministic while the LLM component is used in a controlled manner mainly for improving explanation quality and readability when required [9], [12]. The inclusion of contextual warnings also helped the system provide more practical guidance to users while interacting with online forms and privacy related content [10], [15].

- Future scope

- **RAG Integration:** The use of Retrieval-Augmented Generation (RAG) this could help the framework generate explanations by referring to more relevant policy content instead of relying only on direct model responses [14].
- **Advanced Multilingual Support:** The multilingual feature can also be improved further in future work. Instead of focusing only on translation the system could better understand legal meanings, regional language differences, and context specific interpretations across different languages [15].
- **Cross-Platform Deployment:** The current version of the framework mainly works through a browser extension. In the future, support can also be added for mobile and desktop applications so that the system becomes easier to access across different devices [10], [15].

VIII. GRAPH

A. Fig. 1 — Risk Score Distribution Across 27 Analyzed Websites.

Fig. 1 shows risk scores for 27 websites analysed by the proposed system. The websites are represented using anonymized IDs ranging from W01 to W27. The scores range from 13 (W26) to 65 (W06 and W13) showing that the system can identify different levels of privacy risk across different legal documents [1], [7]. Two websites fell under the High risk category both receiving scores of 65. Medium risk was identified in six websites, with scores between 35 and 42. Most of the remaining websites were grouped in the Low risk category and many of these scores stayed close to 14. The dashed lines at 30 and 60 mark the boundaries between risk categories used by the system [4], [15]. Several Low risk websites had a score of 14 suggesting that these documents share similar lower-risk clauses, such as limits on legal responsibility or collection of user data. These clauses were identified with similar confidence levels [2], [4]. The results show that the rule-based scoring process produces stable and repeatable results across different document types [2], [13].

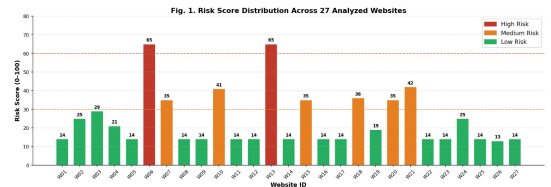


Fig. 1. Distribution of Risk Categories Across 27 Analyzed Websites

B. Fig. 2 shows the distribution of risk categories across 27 websites.

A large portion of the websites fell under the Low risk category, with 19 out of 27 websites (70.4%). Medium

risk was identified in 6 websites (22.2%) whereas only 2 websites (7.4%) were classified as High risk. Most websites were classified as Low risk suggesting that many of the analysed services include common privacy practices that do not activate higher-risk clause patterns in the system [1], [7]. About (29.6%) of the analysed websites were classified as Medium or High risk. This suggests that nearly one in three websites may contain clauses that could create privacy concerns for users, including third-party data sharing without restrictions or limits on legal responsibility [1], [15]. This distribution also shows that the system avoids excessive classification, which can sometimes occur in fully LLM-based systems [9], [12]. In such systems, changes in generated outputs may lead to inconsistent risk assignments.

Fig. 2. Distribution of Risk Categories Across 27 Analyzed Websites

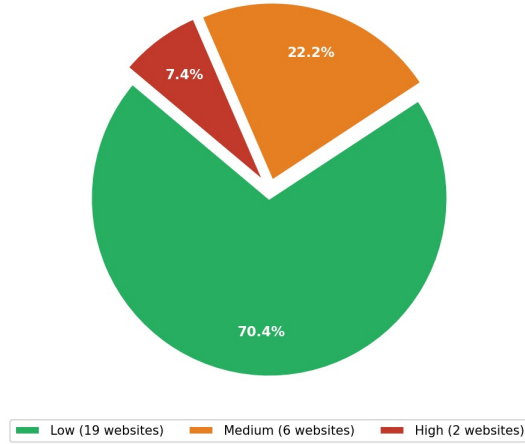
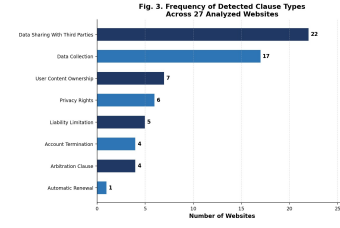


Fig. 2. Distribution of Risk Categories Across 27 Analyzed Websites

C. Fig. 3 — Frequency of Detected Clause Types Across 27 Analyzed Websites

Fig. 3 shows how often different clause types were detected across the 27 examined websites arranged from highest to lowest frequency. Data Sharing With Third Parties was found most often appearing in 22 of the 27 websites. Data Collection was the next most common clause type and appeared in 17 websites. These findings are similar to earlier studies showing that third-party data sharing and personal data collection are among the most common privacy-related practices found in online legal documents [1], [7]. User Content Ownership and Privacy Rights were found in 7 and 6 websites. Limits on legal responsibility, account closure, and dispute resolution clauses were found in about 4 to 5 websites [2]. Automatic Renewal was the least common clause and appeared in only 1 website. The system was able to effectively identify different legally important clauses across documents with different formats and levels of complexity [6], [7]. The large number of data-sharing



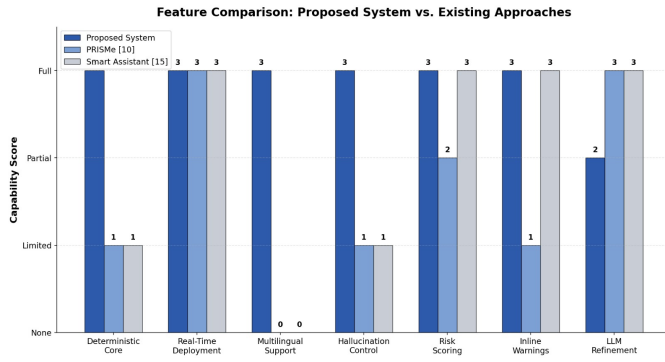
clauses suggests that the proposed system may help users identify common privacy risks while browsing online [10], [15].

D. Fig. 4 — Feature Comparison: Proposed System vs. Existing Approaches

Fig. 4 compares the proposed system with two related systems, PRISMe [10] and Smart Privacy Policy Assistant [15], across seven features. The scores are assigned qualitatively based on the capabilities reported in their published work. These scores are used to provide a clear comparison of system features and are not intended as a quantitative evaluation. The proposed system shows Full capability in Deterministic Core, Real-Time Deployment, Multilingual Support, Hallucination Control, Risk Scoring, and Inline Warnings [2], [13]. LLM Refinement receives a Partial score because the system uses LLMs only when needed rather than relying on them for the main processing task [12], [13]. PRISMe [10] performs well in Real-Time Deployment and LLM Refinement. At the same time, it shows limited support in Deterministic Core, Hallucination Control, and Inline Warnings. The system provides no explicit multilingual support. Earlier evaluations of the system also reported issues related to consistency and user trust because LLM responses may vary [9], [10]. Smart Privacy Policy Assistant [15] shows Full capability in Real-Time Deployment, Risk Scoring, Inline Warnings, and LLM Refinement. However, it remains Limited in Deterministic Core, Hallucination Control, and Multilingual Support. This is mainly because the system depends on LLM-generated outputs for core classification tasks [15]. The comparison shows that the proposed system provides a wider and more balanced set of features, especially in areas related to reliable outputs and privacy-sensitive deployment [13], [14].

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